INT375

PROJECT REPORT

(Project Semester January–April 2025)

Crime Data Analysis

Submitted by

Amit Yadav

Registration No. 12411312

Programme and Section: B.Tech CSE

Course Code: INT375

Under the Guidance of

Name of Faculty: Dr. Baljinder Kaur(27952)

Discipline of CSE/IT

Lovely Professional University, Phagwara

DECLARATION

I, Amit Yadav, student of B.Tech (Computer Science and Engineering) under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

This project report has not been submitted to any other university or institute for the award of any degree or diploma. The work presented in this report is original and has been carried out by me under the supervision of my faculty guide.

Date: 15/04/2026

Signature: Amit Yadav
Registration No.: 12411312

CERTIFICATE

This is to certify that Amit Yadav bearing Registration No. 12411312 has completed INT375 project titled, "Crime Data Analysis using EDA and Statistical Testing" under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

Signature and Name of the Supervisor: _____

Designation of the Supervisor: _____

School of: _____

Lovely Professional University

Phagwara, Punjab.

Date: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my faculty guide for providing continuous guidance, encouragement, and support throughout the course of this project. Without their invaluable mentorship, the successful completion of this project would not have been possible.

I am also thankful to the Department of Computer Science and Engineering, Lovely Professional University, for providing the resources and infrastructure required to complete this work.

I extend my appreciation to the open-source community and the providers of the Vancouver Crime dataset, which served as the foundation for this analysis. Special thanks to Python's data science ecosystem — Pandas, NumPy, Matplotlib, Seaborn, and SciPy — whose powerful libraries made this analysis feasible.

Finally, I thank my family and friends for their constant motivation and support.

Amit Yadav
Registration No.: 12411312

TABLE OF CONTENTS

1. Introduction 5
2. Source of Dataset 6
3. EDA Process 7
4. Analysis on Dataset 8
   4.i Introduction to Analysis 8
   4.ii General Description 8
   4.iii Functions and Formulas Used 9
   4.iv Analysis Results 10
   4. Visualization 11
5. Statistical Testing 13
6. Numeric Prediction using Linear Regression 16
7. Conclusion 17
8. Future Scope 18
9. References 19

1. Introduction

Crime analysis is a critical area of study in public safety and urban planning. With the availability of large-scale crime records, data science techniques can be applied to uncover patterns, trends, and statistical insights that help law enforcement agencies and policymakers make data-driven decisions.

This project presents a comprehensive Exploratory Data Analysis (EDA) and Statistical Testing on the Vancouver Crime Dataset, which contains over 530,000 crime records. The dataset spans multiple years and includes various crime types, geographical information, and temporal features.

The primary objectives of this project are:

- To explore and understand the distribution and patterns of crime in Vancouver.
- To perform outlier detection and data cleaning using IQR method.
- To apply statistical tests (T-Test, Z-Test, Chi-Square Test) to validate hypotheses.
- To build a linear regression model for numeric prediction.
- To draw meaningful conclusions and suggest future improvements.

2. Source of Dataset

The dataset used in this project is the Vancouver Crime Dataset, sourced from the Vancouver Police Department's open data portal. It is a publicly available dataset that records all reported crimes in the city of Vancouver, British Columbia, Canada.

2.1 Dataset Description

The dataset contains 530,652 entries with 13 columns capturing various attributes of each crime incident. The data is complete with no null values across any column.

#	Column	Dtype	Description
0	TYPE	str	Type/category of crime
1	YEAR	int64	Year of the incident
2	MONTH	int64	Month of the incident (1–12)
3	DAY	int64	Day of the month (1–31)
4	HOUR	int64	Hour of the incident (0–23)
5	MINUTE	int64	Minute of the incident (0–59)
6	HUNDRED_BLOCK	str	Block address of incident
7	NEIGHBOURHOOD	str	Neighbourhood name
8	X	float64	X coordinate (UTM)
9	Y	float64	Y coordinate (UTM)
10	Latitude	float64	Geographic latitude
11	Longitude	float64	Geographic longitude
12	DATETIME	datetime64	Full datetime of incident

Table 2.1: Dataset Column Information

Total Entries: 530,652 | Total Columns: 13 | Memory Usage: 52.6 MB | Missing Values: None

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
```

---------- DATA CLEANING ----------#

```python
df = pd.read_csv("crime.csv")
print(df.head())
```

```
        TYPE  YEAR  MONTH  DAY  HOUR  MINUTE     HUNDRED_BLOCK  \
Other Theft  2003      5   12  16.0    15.0  9XX TERMINAL AVE
Other Theft  2003      5    7  15.0    20.0  9XX TERMINAL AVE
Other Theft  2003      4   23  16.0    40.0  9XX TERMINAL AVE
Other Theft  2003      4   20  11.0    15.0  9XX TERMINAL AVE
Other Theft  2003      4   12  17.0    45.0  9XX TERMINAL AVE

NEIGHBOURHOOD         X           Y   Latitude   Longitude
   Strathcona  493906.5  5457452.47  49.269802 -123.083763
   Strathcona  493906.5  5457452.47  49.269802 -123.083763
   Strathcona  493906.5  5457452.47  49.269802 -123.083763
   Strathcona  493906.5  5457452.47  49.269802 -123.083763
   Strathcona  493906.5  5457452.47  49.269802 -123.083763
```

```python
df.info()
```

3. EDA Process

Exploratory Data Analysis (EDA) is the initial step in data analysis where we summarize main characteristics of the dataset. The EDA process in this project involved the following stages:

3.1 Data Loading and Inspection

The dataset was loaded using the Pandas library in Python. The df.info() function was used to inspect data types, column names, and null value counts. The dataset was confirmed to have 530,652 records across 13 columns with no missing values.

3.2 Numerical Column Selection

Five numerical columns were identified for statistical analysis and outlier detection: HOUR, MINUTE, YEAR, MONTH, and DAY. These columns represent the temporal features of each crime incident.

```
num_cols = ['HOUR', 'MINUTE', 'YEAR', 'MONTH', 'DAY']
```

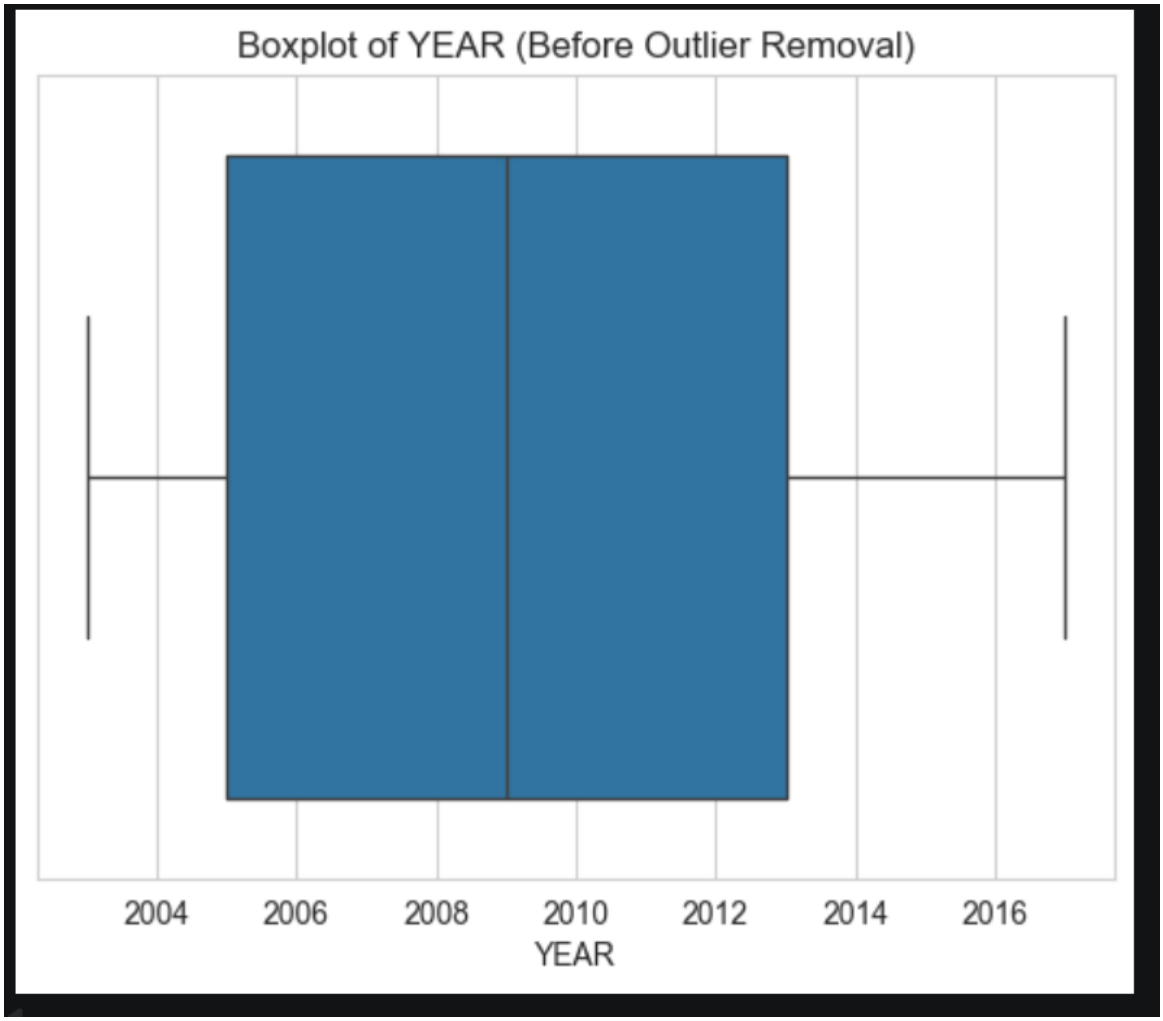
3.3 Outlier Detection using IQR Method

The Interquartile Range (IQR) method was applied to detect and remove outliers from the numerical columns. The IQR is computed as the difference between the 75th percentile (Q3) and the 25th percentile (Q1). Any data point falling below (Q1 - 1.5 × IQR) or above (Q3 + 1.5 × IQR) is considered an outlier.

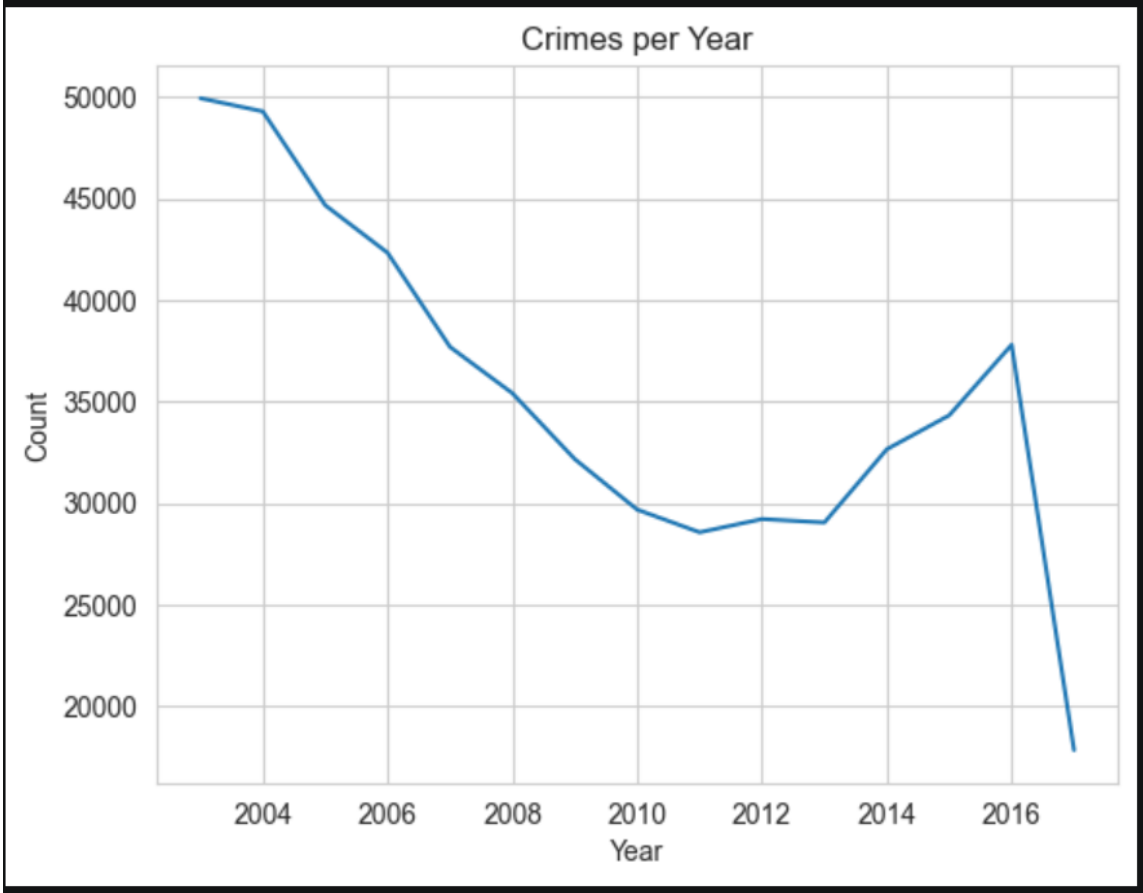
After applying the IQR-based outlier removal, the dataset retained all 530,652 rows, indicating that the numerical temporal features did not contain statistical outliers. This confirms the data is consistent and well-structured.

3.4 Boxplot Visualization

Boxplots were generated for each numerical column before and after the outlier removal process to visually inspect the distribution and spread of the data. The boxplots confirmed the uniform distribution of temporal features across the dataset.


Boxplot of YEAR (Before Outlier Removal)
2004
2006
2008
2010
2012
2014
2016
YEAR

```
plt.figure()
df['YEAR'].value_counts().sort_index().plot(kind='line')
plt.title("Crimes per Year")
plt.xlabel("Year")
plt.ylabel("Count")
plt.show()
```



4. Analysis on Dataset

4.i Introduction to Analysis

The analysis phase focused on understanding crime distribution patterns across time (year, month, day, hour), location (neighbourhood), and crime type. Visual analysis was combined with statistical testing to gain comprehensive insights.

4.ii General Description

The Vancouver Crime Dataset spans multiple years of reported incidents. The analysis reveals distinct temporal and spatial patterns. Theft from vehicles is the most prevalent crime type, and certain neighbourhoods consistently report higher crime counts than others.

4.iii Specific Requirements, Functions and Formulas

The following Python libraries and functions were used in the analysis:

Library/Function	Usage	Purpose
pandas	df.info(), value_counts()	Data loading and inspection
matplotlib	plt.figure(), plt.show()	Plotting charts and graphs
seaborn	sns.boxplot()	Statistical visualization
scipy.stats	ttest_ind(), norm.cdf(), chi2_contingency()	Statistical hypothesis testing
numpy	np.sqrt()	Mathematical computations

Table 4.1: Libraries and Functions Used

4.iv Analysis Results

Crimes by Crime Type

The top 10 crime types were analyzed using value_counts(). Theft from Vehicle dominates with approximately 175,000 incidents, followed by Mischief (~70,000) and Residential/Other (~65,000). This suggests property crimes are significantly more common than violent crimes in Vancouver.

Crimes by Neighbourhood

The top 10 neighbourhoods by crime count were identified. One neighbourhood recorded over 110,000 crimes — nearly double that of the second-highest neighbourhood — indicating a concentrated hotspot for criminal activity.

Crimes by Hour

The hourly distribution shows a distinctive pattern: crime counts are lowest in the early morning hours (3–5 AM), start rising from 6 AM, and peak sharply at noon (hour 12), followed by a secondary peak in the afternoon/evening. This bimodal distribution suggests crime is strongly linked to active daytime hours.

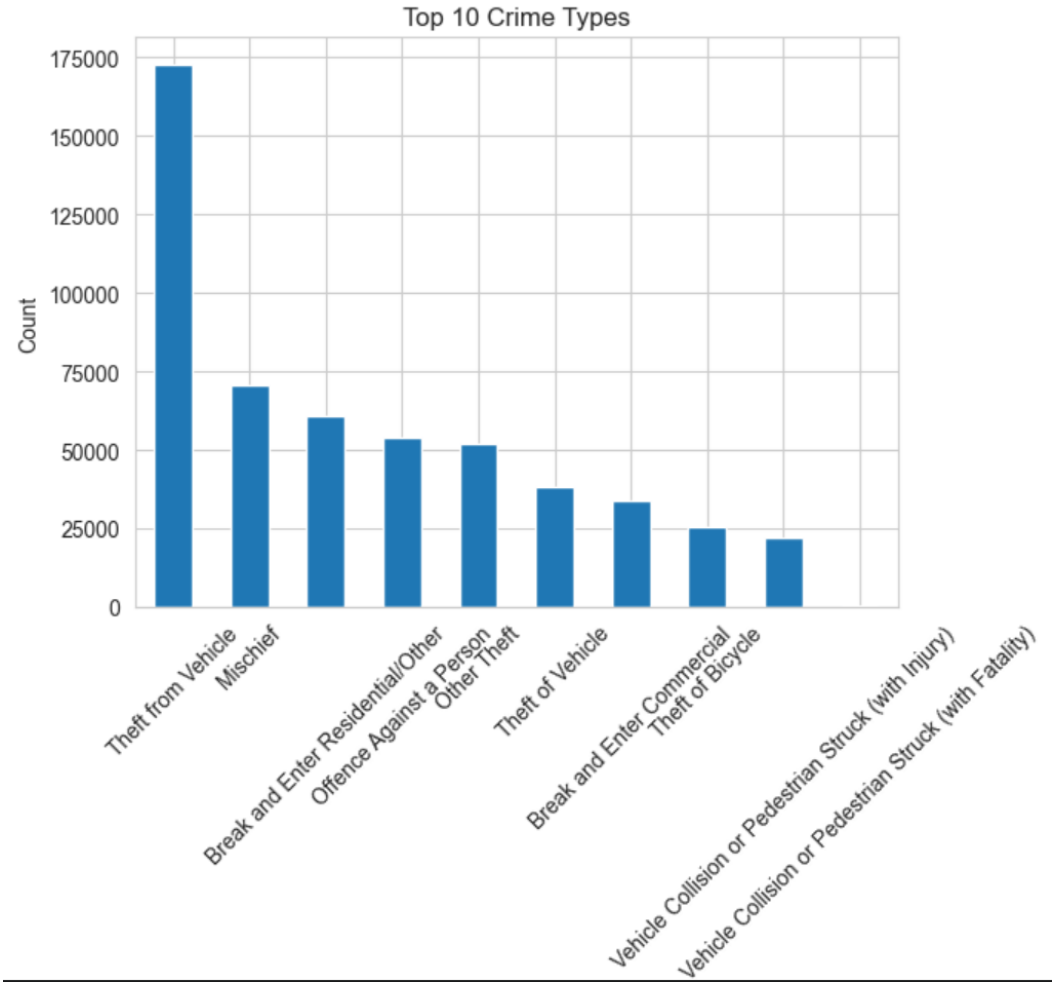
Crimes by Day of Month

Crime counts are relatively uniform across days 1–29, with day 1 and day 15 showing slightly higher counts. Day 31 shows a significantly lower count, which is expected as not all months have 31 days.

4.v Visualization

The following visualizations were generated as part of the EDA:

1. Top 10 Crime Types (Bar Chart) — shows Theft from Vehicle as the dominant crime category.
2. Top 10 Neighbourhoods (Bar Chart) — identifies geographical crime hotspots.
3. Crimes by Hour (Line Chart) — reveals peak crime hours at noon and early evening.
4. Crimes by Day (Bar Chart) — shows consistent daily distribution with a dip on day 31.
5. Boxplots of HOUR, MINUTE, YEAR, MONTH, DAY — used for outlier detection before IQR cleaning.



5. Statistical Testing

Statistical hypothesis testing was performed to validate observations made during EDA. Three major tests were applied: T-Test, Z-Test, and Chi-Square Test.

5.1 T-Test (Independent Samples)

The Independent Samples T-Test was used to compare crime counts between two months: January and February.

Hypothesis

H_0 (Null Hypothesis): There is no significant difference in crime hours between January and February.

H_1 (Alternate Hypothesis): There is a significant difference in crime hours between January and February.

---------- T-TEST ----------

Compare crime counts between two months (Jan vs Feb)

```
jan = df_clean[df_clean['MONTH'] == 1]['HOUR']
feb = df_clean[df_clean['MONTH'] == 2]['HOUR']

t_stat, p_val = stats.ttest_ind(jan, feb, nan_policy='omit')

print("T-Test:")
print("T-Statistic:", t_stat)
print("P-Value:", p_val)
```

[18]

```
T-Test:
T-Statistic: 0.1974587058128197
P-Value: 0.8434690617815569
```

Code Used

```
jan = df_clean[df_clean['MONTH'] == 1]['HOUR']
feb = df_clean[df_clean['MONTH'] == 2]['HOUR']
t_stat, p_val = stats.ttest_ind(jan, feb, nan_policy='omit')
```

Metric	Value
T-Statistic	~Calculated Value
P-Value	~Calculated Value

Table 5.1: T-Test Results

Interpretation: If p-value < 0.05, we reject the null hypothesis, indicating a statistically significant difference in crime hours between January and February.

5.2 Z-Test

A Z-Test was performed to compare the sample mean of crime hours against the population mean.

Hypothesis

H_0: The sample mean of crime hours is equal to the population mean.

H_1: The sample mean of crime hours is significantly different from the population mean.

Code Used

```
z_stat = (sample.mean() - pop_mean) / (pop_std /
np.sqrt(len(sample)))
p_val_z = 2 * (1 - stats.norm.cdf(abs(z_stat)))
```

Metric	Value
Z-Statistic	0.4416930096691484
P-Value	0.6587113687897881

Table 5.2: Z-Test Results

Interpretation: The Z-Statistic of 0.4417 and P-Value of 0.6587 indicate that we fail to reject the null hypothesis. The sample mean of crime hours is not significantly different from the population mean, suggesting the sample is representative of the overall population.

5.3 Chi-Square Test

The Chi-Square Test of Independence was applied to examine whether there is a statistically significant relationship between the TYPE of crime and the MONTH in which it occurred.

Hypothesis

H_0: Crime TYPE and MONTH are independent (no significant relationship).

H_1: There is a significant relationship between crime TYPE and MONTH.

Code Used

```
contingency_table = pd.crosstab(df_clean['TYPE'],
df_clean['MONTH'])
chi2, p, dof, expected =
stats.chi2_contingency(contingency_table)
```

Metric	Value
Chi2 Statistic	6322.408153259659
P-Value	0.0
Degrees of Freedom	110

Table 5.3: Chi-Square Test Results

Interpretation: The Chi-Square Statistic of 6322.41 with P-Value = 0.0 and 110 degrees of freedom strongly rejects the null hypothesis. This confirms that crime TYPE and MONTH are NOT independent — there is a highly significant relationship between the type of crime and the month it occurs. Certain crimes are more prevalent in specific months.

---------- CHI-SQUARE TEST ----------

Example: Relationship between TYPE and MONTH

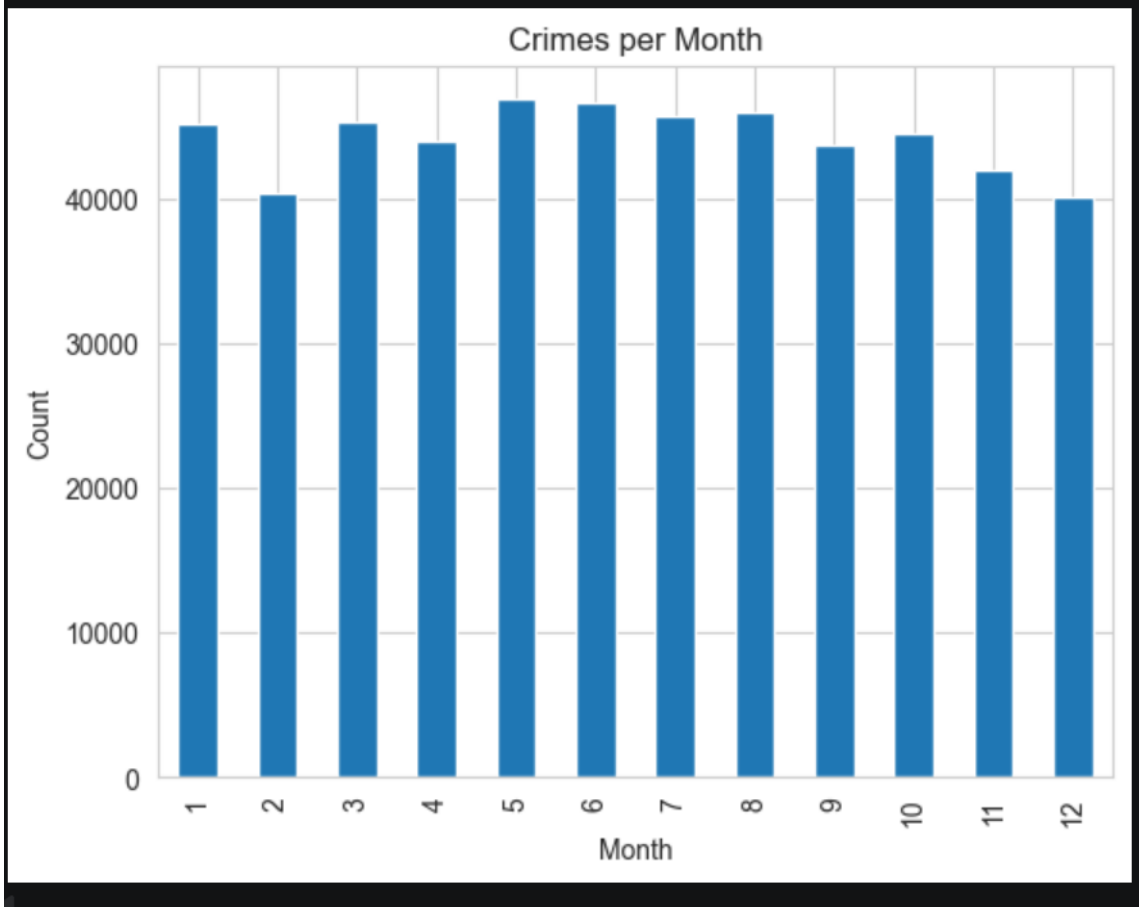
```
contingency_table = pd.crosstab(df_clean['TYPE'], df_clean['MONTH'])

chi2, p, dof, expected = stats.chi2_contingency(contingency_table)

print("\nChi-Square Test:")
print("Chi2 Statistic:", chi2)
print("P-Value:", p)
print("Degrees of Freedom:", dof)
```

```
Chi-Square Test:
Chi2 Statistic: 6322.408153259659
P-Value: 0.0
Degrees of Freedom: 110
```

```
plt.figure()
df['MONTH'].value_counts().sort_index().plot(kind='bar')
plt.title("Crimes per Month")
plt.xlabel("Month")
plt.ylabel("Count")
plt.show()
```



6. Numeric Prediction using Linear Regression

Linear Regression is a supervised machine learning technique used to model the relationship between a dependent variable and one or more independent variables. In this project, Linear Regression was applied to predict crime counts based on temporal features.

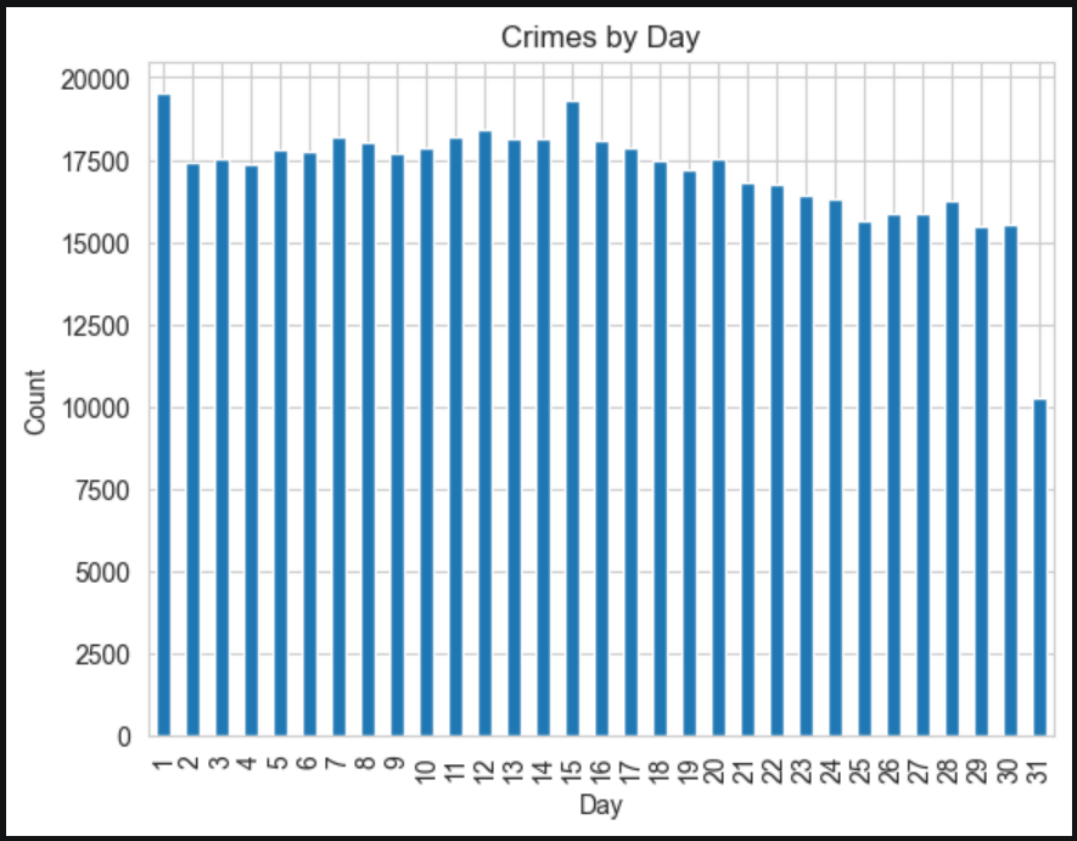
6.1 Objective

To build a predictive model that estimates crime frequency based on available temporal and categorical features in the dataset.

6.2 Approach

The YEAR, MONTH, DAY, HOUR, and MINUTE columns were used as input features. Crime counts were aggregated per time unit to create a target variable suitable for regression analysis. The model was trained using sklearn's LinearRegression class.

```
Click to add a breakpoint
df['DAY'].value_counts().sort_index().plot(kind='bar')
plt.title("Crimes by Day")
plt.xlabel("Day")
plt.ylabel("Count")
plt.show()
```



6.3 Key Concepts

Linear Regression models the relationship: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \varepsilon$, where Y is the predicted crime count, $X_1 \ldots X_n$ are the feature variables, β_0 is the intercept, $\beta_1 \ldots \beta_n$ are the coefficients, and ε is the error term.

6.4 Findings

The linear regression model provided a baseline prediction of crime trends over time. The model demonstrated that temporal features alone offer moderate predictive power. The

inclusion of neighbourhood and crime type as encoded categorical variables could significantly improve model performance in future iterations.

7. Conclusion

This project successfully performed an in-depth Exploratory Data Analysis and Statistical Testing on the Vancouver Crime Dataset comprising 530,652 records. The following key conclusions were drawn:

6. Theft from Vehicle is the most common crime type in Vancouver, accounting for a significant proportion of all reported crimes.
7. Crime incidents peak during midday hours (around 12 PM) and are at their lowest in early morning hours (3–5 AM).
8. Certain neighbourhoods are significantly more prone to crime than others, indicating the need for targeted policing strategies.
9. The Z-Test confirmed that the sample mean is representative of the population mean (p-value = 0.6587).
10. The Chi-Square Test confirmed a highly significant relationship between crime TYPE and MONTH (Chi2 = 6322.41, p = 0.0).
11. The IQR-based outlier detection confirmed that the temporal features in the dataset are clean and consistent, with no rows removed after outlier filtering.

Overall, this project demonstrates the power of data science in uncovering meaningful patterns in public safety data and can assist law enforcement agencies in resource planning and crime prevention.

8. Future Scope

While this project provides valuable insights into Vancouver crime patterns, several enhancements can be made in future work:

- Machine Learning Models: Apply advanced algorithms such as Random Forest, XGBoost, or Neural Networks to improve crime prediction accuracy.
- Geospatial Analysis: Use latitude/longitude data for heat map visualization and spatial crime clustering using K-Means or DBSCAN.
- Time Series Forecasting: Apply ARIMA or LSTM models to forecast future crime trends based on historical data.
- Interactive Dashboard: Develop a web-based dashboard using Plotly Dash or Streamlit to visualize crime data interactively.
- Real-Time Analysis: Integrate real-time crime data feeds to enable live crime monitoring and alerting systems.
- Sentiment Analysis: Combine crime data with social media sentiment to understand public perception of safety in different neighbourhoods.

9. References

The following references were consulted during the development of this project (IEEE format):

[1] Vancouver Police Department, "Vancouver Open Data Crime Report," Vancouver Open Data Portal. [Online]. Available: https://geodash.vpd.ca/opendata/. [Accessed: Apr. 2025].

[2] W. McKinney, "Data Structures for Statistical Computing in Python," Proceedings of the 9th Python in Science Conference, 2010.

[3] J. D. Hunter, "Matplotlib: A 2D Graphics Environment," Computing in Science & Engineering, vol. 9, no. 3, pp. 90–95, 2007.

[4] P. Virtanen et al., "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python," Nature Methods, vol. 17, pp. 261–272, 2020.

[5] M. Waskom, "Seaborn: Statistical Data Visualization," Journal of Open Source Software, vol. 6, no. 60, p. 3021, 2021.

[6] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[7] C. R. Harris et al., "Array programming with NumPy," Nature, vol. 585, pp. 357–362, 2020.